

SHARIF
UNIVERSITY OF TECHNOLOGY



Energy Conversion

Assignment 7

Aria Fateh Kerdari

July 1, 2026

1 Problem 1

At no load and neglecting armature resistance,

$$E_A = V_T = 12 \text{ V}$$

Using the DC machine emf equation,

$$E_A = \frac{ZP}{60a} \phi n_m$$

Hence,

$$n_m = \frac{60aE_A}{ZP\phi} = \frac{60 \times 2 \times 12}{64 \times 2 \times 0.013} = 865.38 \text{ r/min}$$

The angular speed is

$$\omega_m = n_m \frac{2\pi}{60} = 90.62 \text{ rad/s}$$

Using the right-hand rule, the left-side conductors experience an upward force while the right-side conductors experience a downward force. These forces produce a **clockwise** torque.

Neglecting armature resistance, all electrical input power is converted into mechanical power:

$$P_{conv} = 55 \text{ W}$$

Using

$$P_{conv} = \tau_{ind} \omega_m,$$

the induced torque is

$$\tau_{ind} = \frac{P_{conv}}{\omega_m} = \frac{55}{90.62} = 0.607 \text{ N} \cdot \text{m}$$